MIL-F-46580C
15 November 1973
SUPERSEDING
MIL-F-46580B
7 May 1968

MILITARY SPECIFICATION

FUZE, POINT DETONATING, M505A3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing of Fuze, Point Detonating (PD), M505A3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1305

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2.1 (Cont)

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

D7258863 - Fuze, PD, M505A3, Assembly
IEL 7258863 - Index of Inspection Equipment Lists for: Fuze, PD, M505A3, Assembly
D7259023 - Cartridge, 20mm, High Explosive Incendiary, M56A3
D7259135 - Cartridge, 20mm, for M505A3 Fuze Safety and Security Test
F7548083 - Carton, Packing, Ammunition, for Fuze, PD, M505 Series, Assembly, Details and Packing
F7548084 - Box, Packing, Ammunition, for Fuze, PD, M505 Series, Assembly, Details and Packing

PUBLICATIONS

U. S. ARMY MATERIEL COMMAND

AMCR 715-505- Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings, and publications, required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The fuze shall comply with the requirements of Drawing (Dwg.) D7258863, referenced specifications and other requirements specified herein.

3.2 First Article Sample.-Unless otherwise directed by the contracting officer, a First Article sample shall be required.

3.3. Stability.

3.3.1 Jolt.-The fuze assembly shall remain safe during and after the jolt test and shall show no evidence of an explosive element being initiated or parts being broken or coming apart. The fuze assembly shall not be affected in such a manner as to make the fuze unsafe to handle, transport, use or store.

3.3.2 Jumble.-The fuze assembly shall remain safe during and after the jumble test and shall show no evidence of an explosive element being initiated or parts being broken or coming apart. The fuze assembly shall not be affected in such a manner as to make the fuze unsafe to handle, transport, use or store.

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3.4 Safety and Security.-The fuze assembly, loaded and assembled into the cartridge specified by Dwg. D7259135, shall not function, separate or break up during uninterrupted flight of the projectile.

3.5 Function.-The fuze assembly, loaded and assembled into the cartridge specified by Dwg. D7259023, shall cause the projectile to function with high order detonation upon impact against the target.

3.6 Workmanship.-The requirements for workmanship are as specified by the applicable drawings, referenced specifications and the following:

3.6.1 Loading and Assembly.-The fuze assembly shall be free of deformities, missing or inverted components, partial explosive charges; missing or insufficient plating, seals, crimps and other forms of substandard workmanship.

3.6.2 Cleanliness.-The fuze assembly and its components shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of Inspections.-The inspection requirements specified herein are classified as follows:

1. First Article Inspection (see 4.3).
2. Quality Conformance Inspection (see 4.4).

4.3 First Article Inspection.

4.3.1 First Article Sample.-At the beginning of production, a sample representative of the production material, equipment, processes and procedures shall be submitted in accordance with contract requirements and shall consist of 125 fuze assemblies. Prior to approval of the first

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article sample, the acquisition of material or components for, or the commencement of production of the balance of the contract quantity shall be at the sole risk of the contractor.

4.3.1.1 First Article Inspection.-Prior to submission of the first article sample to the Government, the contractor shall inspect all components and assemblies for all contract, drawings, and specification requirements. The contractor's inspection shall be subjected to verification by the Government representative.

4.3.1.2 Examination and Tests.-The first article sample will be inspected for all requirements of the drawings and specifications at a Government laborstory or such facility specified in the contract. Determination as to approval of any first article sample shall be based only upon results of initial examinations and tests.

4.3.1.3 First Article Sample Failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.4 Quality Conformance Inspection.

4.4.1 Submission of Product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.4.1.1 Lot.-A lot shall consist of fuze assemblies loaded by one manufacturer in accordance with the contract requirements and the same drawings and drawing revisions and same specification and specification revision. The fuze lot shall contain detonators from one manufacturer and one detonator lot interfix number.

4.4.1.2 Lot Identification.-Lot identification of fuze assemblies shall comply with the requirements of MIL-STD-1168 and the applicable drawings, supplemented as directed by the procuring activity.

4.4.2 Examination.-One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Each subassembly and each fuze assembly found to be defective shall be rejected.

4.4.2.1 Sampling and Acceptance Criteria.-The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the

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4.4.2.1 (Cont)

assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.4.2.1.1.c) In addition, where three or less defects appear in a classification, or where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defects. (see 6.3)

4.4.2.1.1 AQL's.-The following AQL's are assigned to major and minor defects:

a. Individual.

Major	0.25%
Minor	0.40%

b. Class basis.

Major	1.50%
Minor	2.50%

c. Individual defects associated with the class basis.

Major	0.40%

4.4.3 Classification of Defects.-The classification of defects shall be as follows: (see 6.4) 1/

4.4.3.1 Rotor and Detonator Assembly (see Dwg. C7258865, covering a detail of Dwg. D7258863).

Categories and Defects	Method of Inspection
Critical:	
1. Exposed explosive	Visual 2/
2. Detonator stake missing	Visual
Major:	
101. Inside diameter, detonator stake	Gage
102. Spherical surface distorted	Visual 3/
103. Detonator missing	Visual
104. Identification color incorrectly located or missing	Visual
105. Foreign matter	Visual 4/
Minor: None defined.	

4.4.3.2 Rotor, Detonator and Rotor Detent Spring Assembly (see Dwg. C7258871, covering a detail of Dwg. D7258863).

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4.4.3.2 (Cont)

Categories and Defects	Method of Inspection
Critical:	
1. Cracked, split or broken rotor detent spring	Visual
Major: None defined.	
Minor: None defined.	

4.4.3.3 Booster Assembly (see Dwg. C7258873, covering a detail of Dwg. D7258863).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Rotor seat distorted	Gage 5/
102. Exposed explosive	Visual 6/
103. Charge missing	Visual 7/
104. Low charge	Visual 8/
105. Crack or split in booster holder	Visual
106. Thread damaged	Visual 9/
107. Foreign matter	Visual
Minor: None defined.	

4.4.3.4 Assembly of Rotor-Detonator and Rotor Detent Spring Assembly in the Metal Parts Assembly.

Categories and Defects	Method of Inspection
Critical:	
1. Rotor detent spring missing	Visual 10/
Major: None defined.	
Minor: None defined.	

4.4.3.5 Fuze Assembly (see Dwg. D7258863).

Categories and Defects	Method of Inspection
Critical: None defined.	

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4.4.3.5 (Cont)

Categories and Defects	Method of Inspection
Major:	
101. Length, booster holder to shoulder	Gage
102. Thread damaged	Visual 9/
103. Protective coating damaged	Visual 11/
104. Nose cap damaged	Visual 12/
105. Loose nose cap	Manual 13/
106. Crack or split in assembly	Visual
107. Foreign matter	Visual
108. Loose booster holder	Manual 13/
Minor:	
201. Sealant missing or faulty	Visual 14/

4.4.3.6 Packed Carton (see Dwg. F7548083).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Carton incorrectly closed or sealed	Visual 15/
102. Label missing	Visual
103. Marking incorrect, incomplete, illegible or missing	Visual
104. Carton damaged	Visual 16/
105. Fuze inverted in separator	Visual
106. Filler, spacer or separator missing	Visual
Minor:	
201. Marking touched up or repaired with non-permissible material (crayon, chalk, etc.)	Visual
202. Numerical content incorrect	Count

4.4.3.7 Sealed Packing Box (see Dwg. F7548084).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Box damaged	Visual 17/
102. Marking incorrect, incomplete, illegible or missing	Visual
103. Filler missing	Visual
104. Strapping loose or missing	Visual

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4.4.3.6 (Cont)

Categories and Defects	Method of Inspection
Minor:	
201. Marking touched up or repaired with non-permissible material (crayon, chalk, etc.)	Visual

1/ MIL-STD-651 shall apply in defining and evaluating visual defects.

2/ Visible explosive trapped between the detonator closing disc and the cup crimp; exposed through a split, crack, rupture, or perforation in the disc or cup, or loose explosive (flake or cake) on any other visible surface of the assembly, shall be cause for classifying the assembly defective.

3/ A deformation of the spherical surface of the assembly resulting from detonator staking or other form of substandard workmanship shall be cause for classifying the assembly defective.

4/ "Foreign matter" shall be interpreted to exclude "exposed explosive" which is classified elsewhere as a critical defect. All other matter (including sealant on the spherical surface or around the flash hole area of the assembly) shall be classed as "foreign matter".

5/ Rotor seat distortion shall be determined by measuring the distance between the underside of the booster holder flange to the top of a 0.37700 - .00005 inch precision ball positioned in the rotor seat.

6/ A missing closing disc or one which is split, cracked, ruptured, torn, perforated or crimped off-set to expose a portion of the explosive, shall be cause for classifying the assembly defective.

7/ The booster holder is found crimped with the closing disc missing or resting on the bottom of the charge cavity.

8/ A concavity of magnitude in the crimped closing disc which indicates that the disc is not supported by the explosive charge, or a crimped assembly which contains a loose closing disc resting on a partial charge, shall be interpreted as a low charge.

9/ The booster assembly or body assembly shall be classed defective if the thread is crushed or otherwise distorted to a degree that prevents acceptance by the "Go" thread gage.

10/ Visual inspection for "rotor detent spring missing" shall be performed on the rotor-detonator and rotor detent spring assembly prior to and after its insertion into the metal parts assembly.

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11/ The fuze assembly shall be classed defective if the plating film on the nose cap or body ogive is scored or otherwise broken to visibly expose basis metal.

12/ The fuze assembly shall be classed defective if the nose cap is cracked, split, perforated, dented or otherwise deformed.

13/ The fuze assembly shall be classed defective if the nose cap is not crimped or turns on the body, or if the booster holder turns in the loosening direction, upon the application of thumb and finger torque.

14/ The assembly shall be classed defective, if the sealant does not seal crimp 360° and cover damaged plating on base of fuze, or if the sealant is present on the fuze body.

15/ A carton which is incorrectly closed or sealed or which lacks sufficient tape or security of seal to adequately protect the contents from hazards normal to handling, interplant shipment or undercover storage, shall be classed defective.

16/ A carton which is torn, punctured or contaminated with oil, grease, moisture or other foreign matter which reduces the protective quality of the carton, shall be classed defective.

17/ A missing, broken or split board or cleat; a broken, missing or malfunctioning hardware component or other defect which weakens the box structure or reduces the protective quality of the box, shall be classed as damage.

4.4.4 Tests.-The tests listed in Table I shall be conducted on each lot in accordance with the test methods and procedures specified by 4.5. The sampling plans for these tests shall be as specified by Table I.

Table I

Test		Sample Size	Item	Rqmt Para
Assembly equipment verification	1/		2-Verification standards (see 4.5.1)	3.6.1
Stability				
Jolt	2/	8	Fuze assemblies	3.3.1
Jumble	3/	8	Fuze assemblies	3.3.2
Safety and security	4/	50	Fuze assemblies	3.4
Function	5/	50	Fuze assemblies	3.5

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1/ Failure of the assembly equipment to reject the simulated critical defect standard and accept the correctly assembled and oriented standard, shall be cause for the rejection of all production inspected by the equipment since the preceding assembly equipment verification test. (see 4.5.1)

2/ Failure of one or more units of the sample to comply with the applicable acceptance criteria shall be cause for rejection of the lot.

3/ Failure of one or more units of the sample to comply with the applicable acceptance criteria shall be cause for rejection of the lot.

4/ Failure of one or more units of the sample to comply with the applicable safety and security requirements shall be cause for rejection of the lot.

5/ Failure of four or more units of the sample to comply with the applicable function requirements shall be cause for rejection of the lot. If two or three units of the sample fail to comply with the function requirements, a second sample consisting of the same number of units as specified for in the first sample, shall be tested. If, in the accumulated samples, five or more units fail to comply with the function requirements, the lot shall be rejected. Also, the lot shall be rejected if in either sample, one or more units function prior to impact of the projectile against the target.

4.4.5 Inspection Equipment.-Index of Inspection Equipment List (IEL) 7258863 identifies the applicable inspection equipment lists indicating the equipment required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to Inspection Equipment List for definition of coding, Dwg. B11075228, a detail of IEL 7258863).

4.5 Test Methods and Procedures.

4.5.1 Assembly Equipment Verification.-This test shall apply only where automated inspection is approved and substituted for the visual inspection requirement specified by 4.4.3.4, Critical 1. One equipment verification standard of Table I shall consist of a metal parts assembly with the rotor-detonator subassembly seated therein to simulate the referenced critical defect. The other standard of Table I shall consist of a metal parts assembly with the rotor, detonator and rotor detent spring assembly correctly seated and oriented therein. Each standard shall be cycled three times through the equipment at the beginning of each shift and at intervals not exceeding four hours thereafter.

4.5.2 Jolt.-The method of test shall be as specified in MIL-STD-331, Test No. 101. To determine compliance with the requirements, the fuze assembly shall be disassembled and examined after the test.

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4.5.3 Jumble.-The method of test shall be as specified in MIL-STD-331, Test No. 102. To determine compliance with the requirements, the fuze assembly shall be disassembled and examined after the test. Damage to the threads or body that would prevent assembly shall not be considered as being defective for use.

4.5.4 Safety and Security.-The method of test shall be as specified in AMCR 715-505, Volume 8, for function and casualty, metal component parts, modified as follows: The test cartridges, at ambient temperature, shall be fired in one burst at a cyclic rate not less than 1400 shots per minute in Gun, Automatic, M39. Observation for premature fuze functioning shall be made from a position suitable for detecting the occurrence of a malfunction.

4.5.5 Function.-The method of test shall be as specified in AMCR 715-505, Volume 8, for projectile function. The target shall be 2024 T3, T4 or T36 sheet aluminum of 0.040 inch nominal thickness, located 200 ± 25 yards from the weapon muzzle, normal to the line of fire. Projectile performance shall be observed from the camera position.

5. PREPARATION FOR DELIVERY

5.1 Intermediate Packaging, Labeling and Marking, Level B.-The fuze lot shall be packaged and labeled as specified by Dwg. F7548083.

5.1.1 Packing and Marking, Level B.-The packaged fuze lot shall be packed and marked as specified by Dwg. F7548084.

6. NOTES

6.1 Ordering Data.-Invitation for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Place of inspection if not at the place of manufacture.

6.1.3 First article sample requirements. (see 3.2, 4.3 and 6.2)

6.1.4 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.5 Instructions for shipment of first article and production lot samples for firing tests.

6.2 First Article Sample.-In order to minimize inspection costs, the firing tests as specified in 4.4.4 will be performed after the sample has been found acceptable for all other requirements. Additional fuzes may be required by the testing facility.

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6.3 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.4 Intermediate Point Inspection.-The classification of defects, among other things, identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality in the end item.

NOTE: Asterisks not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - MU
Air Force - 70

Preparing Activity:

Army - MU

Review Activities:

Army - MU
Air Force - 70

Project No. 1905-0419

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